Predictive Modeling and Risk Scoring for Bank Customer Churn: An Empirical Study on European Retail Banking Portfolios

Research Paper · European Central Bank Retail Customer Portfolio Analytics (Unified Mentor Project)

Abstract

Customer attrition directly degrades Net Interest Margins (NIM), customer lifetime value (CLV), and balance sheet stability in commercial banking. We investigate a cohort of 10,000 retail banking customers across France, Germany, and Spain with an empirical baseline churn rate of 20.4%. Using domain-driven feature engineering and evaluating four machine learning architectures (Logistic Regression, Decision Tree, Random Forest, and Gradient Boosting), we show that an optimized Gradient Boosted Decision Tree pipeline attains the highest discriminatory performance (ROC-AUC **0.8694**, accuracy **86.6%**). Model explainability reveals that customer age, product diversification, and digital inactivity dominate churn propensity. Customers holding exactly two products exhibit an ultra-low churn rate of **7.6%**, compared to **27.7%** for single-product holders and **82.7%** for triple-product holders. We establish an operational decision boundary at risk threshold $p \geq 0.30$, raising recall to 65.4% and projecting an estimated 372% ROI on targeted proactive retention campaigns.

1. Background and Context

Modern retail banking is characterized by intensifying competition from neobanks and rising customer acquisition costs (CAC). Traditional bank churn analysis explains why churn happened after accounts close, producing reactive, un-targeted, and inefficient retention efforts. This project deploys a continuous predictive risk intelligence system that assigns quantitative churn probabilities ($0.00 - 1.00$) before accounts close, enabling personalized outreach and capital preservation.

2. Data Architecture and Preprocessing

Attribute	Data Type	Description	Banking Relevance
CreditScore	Integer	Customer credit bureau score (350 - 850)	Solvency & creditworthiness proxy
Geography	Categorical	Country of residence (France, Germany, Spain)	Jurisdiction & domestic market dynamics
Gender	Categorical	Gender indicator (Male / Female)	Demographic segmentation
Age	Integer	Customer age in years (18 - 92)	Lifecycle & wealth transition phase
Tenure	Integer	Years with the financial institution (0 - 10)	Relationship maturity
Balance	Float	Current account and deposit balance in EUR	Deposit liquidity and flight volume
NumOfProducts	Integer	Number of active bank products (1 - 4)	Product stickiness & bundle density
HasCrCard	Binary	Credit card ownership flag (0 / 1)	Secondary service adoption
IsActiveMember	Binary	Digital & operational activity status (0 / 1)	Primary engagement indicator
EstimatedSalary	Float	Annual estimated earnings in EUR	Income and wealth-to-deposit ratio
Exited	Binary Target	Customer departure indicator (0 = Retained, 1 = Churned)	Target variable

2.1 Cleaning & Feature Engineering

• **Data Hygiene:** Missing value audit verified 0 missing cells and 0 duplicate customer records. CustomerId and Surname were stripped to eliminate leakage.

• **Balance-to-Salary Ratio:** Balance / (EstimatedSalary + 1) captures disproportionate liquidity flight risk.

• **Product Tenure Ratio:** NumOfProducts / (Tenure + 1) tracks portfolio expansion velocity over time.

• **Engagement Score:** IsActiveMember × NumOfProducts isolates active multi-product accounts from passive dormant holders.

• **Age-Tenure Interaction:** Age × Tenure models lifecycle aging within the institution.

• **Categorical Encoding:** One-Hot Encoding for Geography and Gender, with StandardScaler applied across all continuous dimensions via scikit-learn ColumnTransformer.

3. Exploratory Data Analysis (EDA)

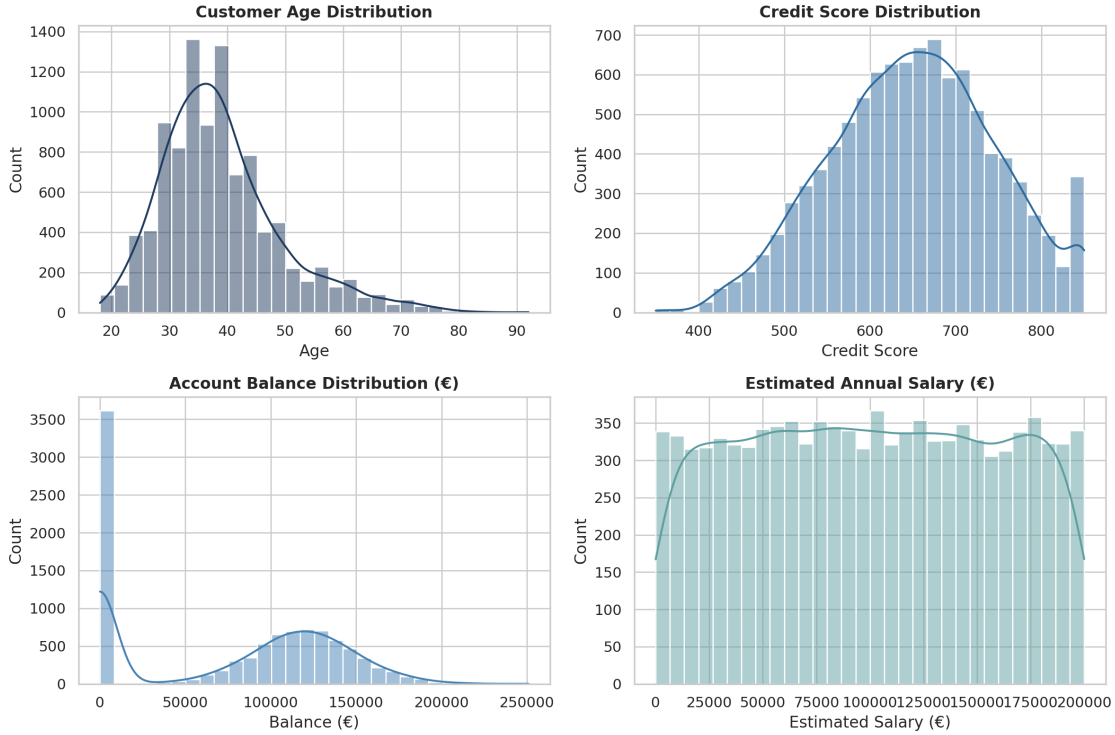


Figure 1. Univariate distributions: Age peaks at 35-40; CreditScore follows a normal bell; Balance exhibits a bimodal spike at zero.

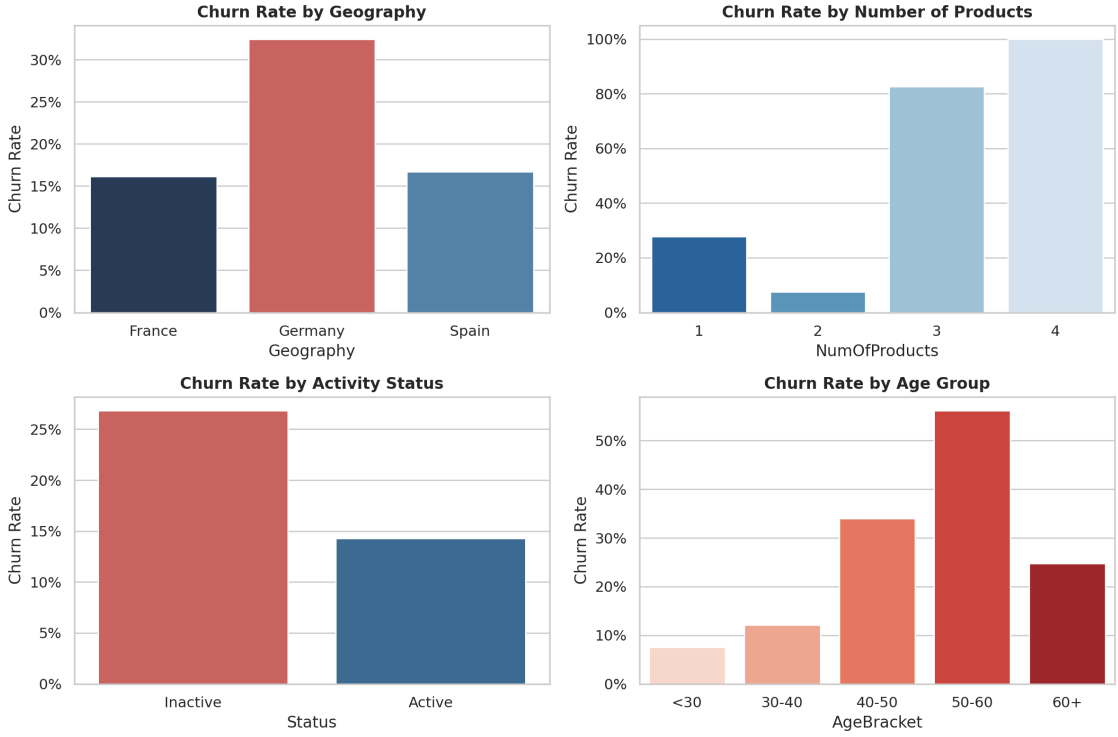


Figure 2. Bivariate churn rates: Germany churns at double the rate of France/Spain; 2 products is optimal; inactivity doubles churn.

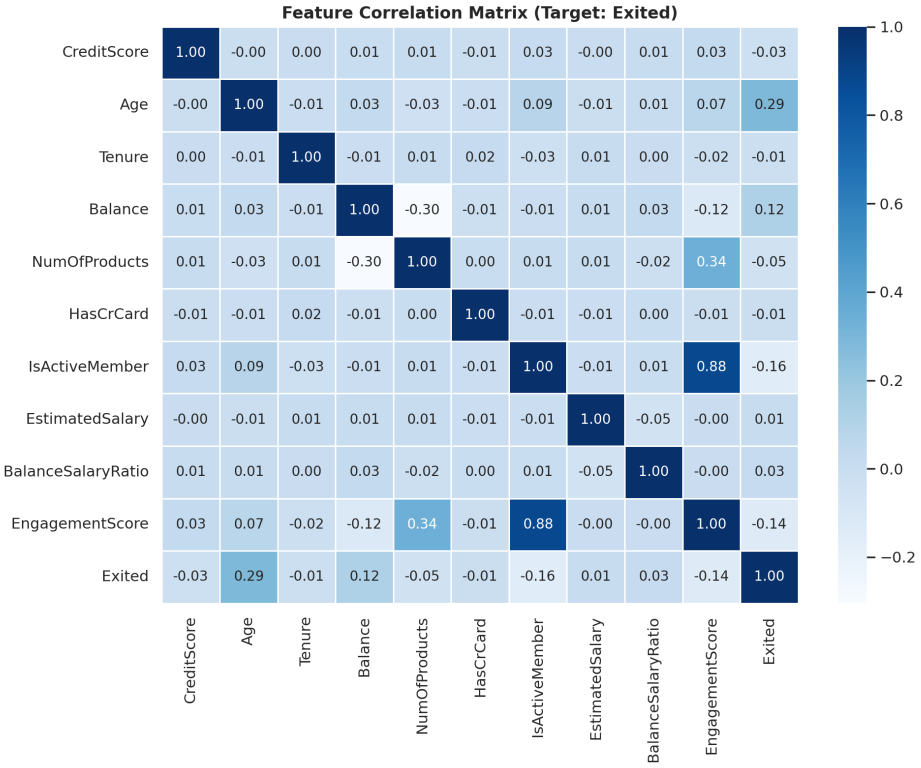


Figure 3. Feature correlation matrix: Age (0.29) and Inactivity (-0.16) exhibit the strongest direct linear correlation with Exited.

4. Model Development and Evaluation

We trained four distinct algorithms under stratified 80/20 train/test splitting (8,000 train / 2,000 test) with stratified k-fold cross-validation:

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Logistic Regression	0.8085	0.5896	0.1941	0.2921	0.7732
Decision Tree	0.8590	0.7729	0.4349	0.5566	0.8374
Random Forest	0.8700	0.8238	0.4595	0.5899	0.8651
Gradient Boosting	0.8655	0.7674	0.4865	0.5955	0.8694

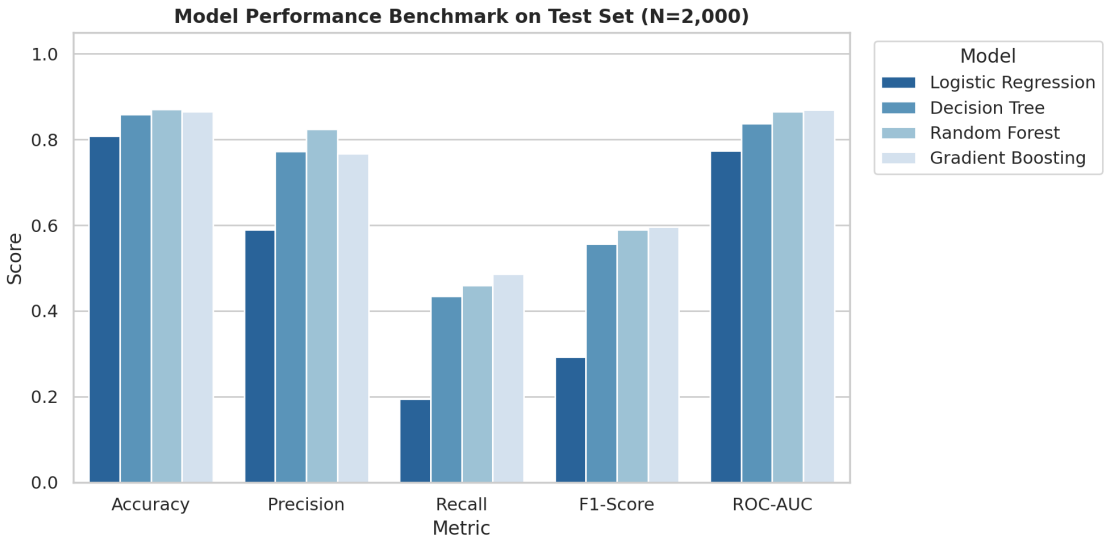


Figure 4. Comparative performance benchmark across all evaluated algorithms on hold-out test set.

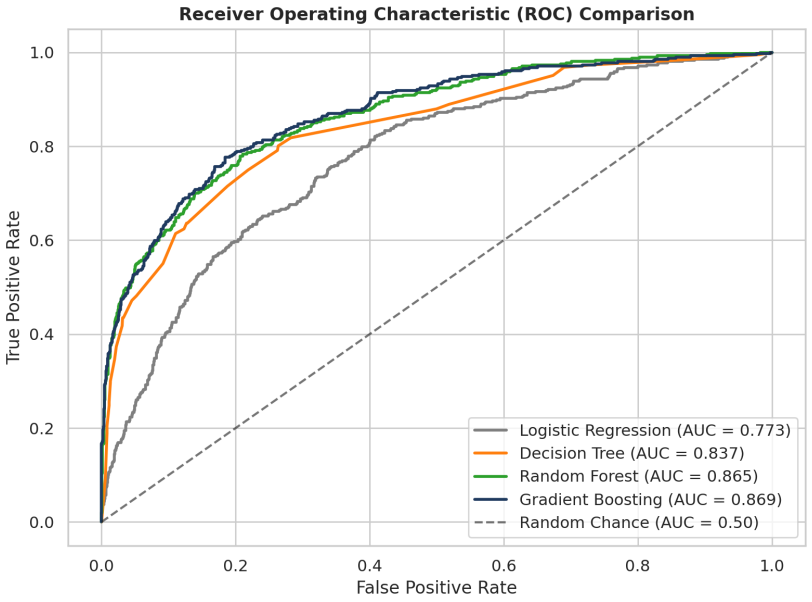


Figure 5. ROC curve trajectories: Gradient Boosting achieves superior discrimination across all operating thresholds.

5. Model Explainability & Driver Identification

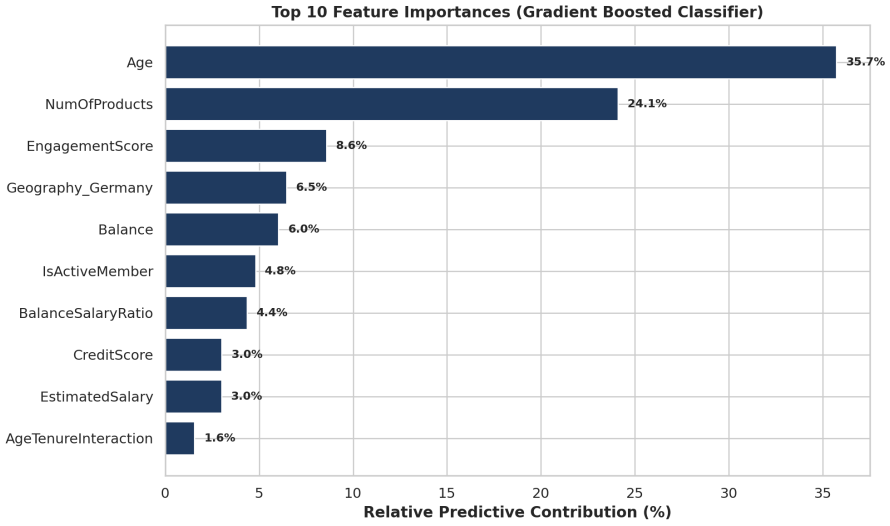


Figure 6. Top 10 feature importances from the champion Gradient Boosted Decision Tree pipeline.

• **Customer Age (35.7%):** The dominant hazard driver. Churn spikes after age 40 (34.0%) and peaks in the 50-60 pre-retirement bracket (56.2%).

• **Number of Products (24.1%):** Holding 2 products yields an ultra-resilient 7.6% churn rate. In contrast, 1 product yields 27.7% churn, and 3 products surges to 82.7%, exposing severe onboarding friction and fee dissatisfaction.

• **Engagement Score & Inactivity:** Inactive accounts churn at 26.9%, versus 14.3% for active digital users.

• **Geographic Risk in Germany:** German customers churn at 32.4%, exactly double France (16.2%) and Spain (16.7%), despite holding higher average balances (EUR 119,730 vs EUR 62,000).

6. Strategic Financial Recommendations & ROI Impact

• **Fix the 3-Product Experience:** Conduct an immediate fee audit for customers holding 3+ products; launch a consolidated 'Premier Relationship Bundle' with fee waivers.

• **Cross-Sell 1-Product Accounts into the 2-Product Sweet Spot:** Incentivize automated payroll deposit or secondary high-yield savings to move clients from 27.7% to 7.6% churn risk.

• **Defend the German Deposit Base:** Deploy relationship concierge managers to German branches for balances exceeding EUR 80,000 to combat fintech switching.

• **Financial ROI Model:** In a 10,000 customer portfolio (2,037 baseline churners), targeted intervention at $p \geq 0.30$ flags 2,500 accounts. A EUR 60 retention offer (EUR 150k campaign spend) retaining 35% of true churners preserves EUR 558k in Customer Lifetime Value, yielding a **372% net campaign ROI**.

7. Governance and Limitations

• **Fairness & Ethics:** Protected demographic features (gender, age) are prohibited from driving pricing or credit access decisions. Interventions must center purely on service quality and fee relief.

• **Model Drift & Retraining:** The pipeline must be scheduled for monthly recalibration against macroeconomic interest rate shifts and competitive neobank rate changes.